Week 7 Assignment Report

Delta Lake Incremental Data Processing using Apache Spark

Project Overview

This report documents the implementation of the Week 7 assignment completed as part of the Celebal Technologies Data Engineering Internship. The assignment focuses on Incremental Data Processing using Apache Spark and Delta Lake while following a structured ETL workflow similar to production environments.

What Was Done

Week 7 Assignment Summary – Delta Lake Incremental Data Processing

This Week 7 assignment was completed with the objective of understanding how modern Data Engineering pipelines process incremental data using Apache Spark and Delta Lake. Instead of treating the assignment as a collection of independent coding tasks, I approached it as a small production-style ETL project, where every step was designed to follow the workflow commonly used in enterprise data platforms.

The project began by loading the Customer Master dataset into Databricks. Rather than directly performing transformations, the dataset was first explored to understand its overall structure. The schema, number of records, column names, and sample data were validated to ensure that the dataset was correctly loaded and ready for processing. This initial exploration helped establish confidence in the source data before moving to any transformation or business logic.

After loading the data, a Data Quality Analysis was performed. Instead of immediately cleaning the dataset, I generated a detailed data quality report that included total records, total columns, null values, duplicate customer IDs, and unique customer counts. This approach reflects a real-world ETL workflow where engineers first assess the quality of incoming data before deciding how to clean it. The report provided a clear understanding of the condition of the source dataset and served as the baseline for all subsequent transformations.

Once the quality assessment was complete, the dataset was cleaned by handling missing values and removing duplicate records wherever required. Rather than modifying the original source directly, a trusted version of the customer data was created, ensuring that all further processing was performed on reliable and validated information. This mirrors the practice followed in production environments, where only trusted datasets are used for downstream pipelines.

The cleaned dataset was then stored as a Delta Table, which became the Customer Master table for the remainder of the assignment. Instead of working only with CSV files, the assignment demonstrated how Delta Lake provides ACID transactions, reliable updates, and efficient incremental processing. This allowed the project to simulate how organizations manage customer master data in enterprise data platforms.

To demonstrate incremental processing, a second dataset was created programmatically instead of manually uploading another file. The incremental dataset represented daily customer changes received from an operational CRM system. It contained two different types of records:

• Existing customers whose information had changed.
• Completely new customers who needed to be inserted into the master table.

Before performing the merge operation, a separate Incremental Change Plan was created to clearly document which customers would be updated and which customers would be inserted. This additional documentation is commonly found in production ETL pipelines because it allows engineers to understand expected changes before modifying the master data.

The core of the assignment was the implementation of Slowly Changing Dimension Type 1 (SCD Type 1) using Delta Lake MERGE. Instead of simply writing the MERGE statement, the notebook first explained the business purpose of SCD Type 1 and when it should be used. Existing customer records were updated with the latest information, while new customer records were inserted into the Delta Table. This demonstrated how Delta Lake efficiently synchronizes only the changed records instead of reprocessing the complete dataset.

After executing the MERGE operation, a dedicated Validation Phase was performed. Multiple validation checks included row count verification, duplicate analysis, validation of updated customer records, verification of newly inserted customers, and overall data integrity checks.

To extend the assignment beyond the mandatory requirements, a separate implementation of Slowly Changing Dimension Type 2 (SCD Type 2) was created. A dedicated Delta Table was built with effective_date, end_date, and is_current columns to preserve historical customer records. A customer update scenario demonstrated how historical information is retained while maintaining the latest version.

Following the implementation of both SCD techniques, a detailed comparison between SCD Type 1 and SCD Type 2 highlighted their practical use cases, storage implications, performance considerations, and business scenarios.

Finally, the entire pipeline was validated through a structured Final Validation Report covering data loading, cleaning, incremental processing, Delta MERGE, SCD Type 1, SCD Type 2, and historical data preservation.

Additional Efforts Beyond Assignment Requirements

Although the assignment mainly required implementing incremental processing using Delta Lake, several additional enhancements were included to make the notebook more structured, practical, and closer to real-world Data Engineering workflows.

• Structured notebook with clear objectives before every major section.
• Business scenarios explaining why each operation is performed in enterprise systems.
• Dedicated Data Quality Report before data cleaning.
• Validation after every important transformation.
• Incremental Change Plan documenting all expected customer updates and inserts before the MERGE operation.
• Production-oriented explanations throughout the notebook.
• Separate implementations of both SCD Type 1 and SCD Type 2.
• Complete validation report verifying the success of the entire pipeline.
• Detailed comparison between SCD Type 1 and SCD Type 2.
• Clean documentation, observations, and project summaries to improve readability and maintainability.

Conclusion

This assignment was completed as a structured Data Engineering project rather than only a coding exercise. The notebook demonstrates the complete lifecycle of an incremental ETL pipeline, from data ingestion and quality analysis to Delta MERGE, validation, and historical data management using SCD Type 2. It emphasizes both implementation and the engineering practices required to build reliable and maintainable data pipelines.